

🔑 Convex Combination

BM25 score

mDPR score

$$s_{CC} = (1 - \alpha) \cdot \tilde{s}_{\text{BM25}} + \alpha \cdot \tilde{s}_{\text{Dense}}$$

Final ranking

🔑 Reciprocal Rank Fusion

BM25 rank r_b

mDPR rank r_d

$$s_{\text{RRF}} = \frac{1}{k + r_b} + \frac{1}{k + r_d}$$

Final ranking